

Data-Driven Storytelling: Bridging Knowledge Graphs, GenAI, and Narrative (DDS 2026)

Pasquale Lisena¹, Maria Angela Pellegrino², Lisa-Yao Gan³, Yihang Zhao⁴,
and Yiwen Xing⁵

¹ EURECOM, Sophia Antipolis, France pasquale.lisena@eurecom.fr

² University of Salerno, Italy mapellegrino@unisa.it

³ Technical University Munich, Germany lisa.gan@tum.de

⁴ King's College London, UK yihang.zhao@kcl.ac.uk

⁵ University of Oxford, UK yiwen.xing@eng.ox.ac.uk

Workshop Type: Novel. **Proposed Length:** half-day.

Abstract. DDS 2026 aims to explore innovative methods for transforming complex datasets into engaging, interpretable narratives. This workshop focuses on the intersection of semantic technologies, artificial intelligence, and narrative design, seeking to make data more relevant, interactive, and shareable. We invite researchers and practitioners from diverse fields—including NLP, knowledge representation, HCI, Information retrieval and creative industries—to discuss frameworks and tools that enhance data engagement through storytelling, games, and art. Key topics include ontological coherence, generative AI, and participatory content creation, with the goal of empowering experts to communicate effectively in domains like education, journalism, and cultural heritage.

Topics

- Knowledge graphs and ontologies for narrative representation and generation
- Linked data and semantic annotation for storytelling
- LLMs and Generative AI for data-driven narratives
- Benchmarking narrative coherence and quality
- Data-driven storytelling in journalism, education, and cultural heritage
- Interactive and participatory content creation (e.g., interactive interfaces, immersive experiences, human-AI collaborative authoring)
- Trustworthiness in automated storytelling (bias, hallucination, transparency)
- Development and evaluation of tools and platforms for data-driven stories
- Human-data interaction and crowdsourcing for narrative design
- Integration of methods from data science, HCI, cognitive psychology, creative writing, and visual arts
- Impact of localization, topicalization, and shareability on public engagement with data

Chairs

Pasquale Lisena (EURECOM, France - mail - homepage) is a researcher in Knowledge Graphs and Information Extraction from natural language. He got a

PhD in Data Science from Sorbonne University in 2019. He took part in research projects like DOREMUS, SILKNOW, Odeuropa, and led the kFLOW project, studying the automatic extraction of relations between events and facts from texts. He served as Poster & Demo chair for TheWebConf 2021, ESWC 2026, and proceedings chair for ESWC 2024 and 2025. He co-organised the workshop Semantic Methods for Events and Stories (SEMMES) at ESWC 2023 and 2024. He is co-editor of the ACM TIST Special Issue on *New Frontiers in Interactive Storytelling and Computational Models of Narrative*.

Maria Angela Pellegrino (University of Salerno, Italy - mail - homepage) is a researcher in Knowledge Graphs, Data Quality, and educational approaches to helping learners understand how to effectively manage data, visualize it, and extract knowledge-driven insights. She earned her PhD in Computer Science from the University of Salerno in 2022. She has significant experience in the community, having served as a workshop organizer for major conferences, including TheWebConf and events in the Semantic Web domain, where she coordinated sessions focused on data quality and knowledge-driven systems. She has chaired conferences and events addressing the intersection of technology, storytelling, and education. She is co-editor of the ACM TIST Special Issue on *New Frontiers in Interactive Storytelling and Computational Models of Narrative*.

Lisa-Yao Gan (Technical University Munich, Germany - mail) is a researcher working at the intersection of information retrieval, generative AI, and knowledge graphs, focusing on transforming complex data into interpretable and trustworthy narratives. Her research explores how generative AI, grounded in structured knowledge graphs, supports dataset discovery, metadata curation, and sense-making in open data ecosystems for data-driven storytelling. She is part of the TUM-IAS Focus Group “*Trustworthy Knowledge Graphs*”. Her work is supported by the Hans Fischer Senior Fellowship awarded to Prof. E. Simperl and funded by Siemens AG. She collaborates closely with public-sector and industry partners, including projects with Data for London on generative AI for metadata curation and with Siemens on KG-based AI risk assessment.

Yihang Zhao (King’s College London, UK - mail - homepage) is a researcher in human-centered AI, studying how UI and interaction techniques of GenAI-based systems can be designed and evaluated to support knowledge workers in collaborative tasks by triggering cognitive and creative thinking process. He is affiliated with the Knowledge Graph Group at King’s College London, led by Prof. E. Simperl and Dr. A. Meroño-Peñuela. He has participated in research projects including MuseIT, ProVe, and PHAWM, and led the development of OntoChat and OntoScope, studying the ontology construction and broader knowledge work. His research is funded by EPSRC, Responsible AI UK, and Siemens AG. He has served as a PC member for CHIWORK 2026, and the ELMKE and LLMs4KG@ESWC workshops since 2026, as well as the Special Track on LLMs for KG at ESWC 2024. He is also a Young Fellow of the Royal Society of Arts.

Yiwen Xing (University of Oxford, UK - mail - homepage) is a researcher in data visualization at the Oxford e-Research Centre, University of Oxford. Her research focuses on the design and evaluation of human-centered visual-

ization and visual analytics systems through collaborative design studies that support sensemaking, decision-making, and knowledge construction in complex, data-intensive workflows. She has conducted cross-domain research in digital humanities, public health forecasting, and explainable AI, contributing to systems and projects including scFlowVis, OwnershipTracker, the Wellcome Trust-funded DART project, the EPSRC-funded RAMPVIS project, and the EPSRC-funded VIS for ML-assisted human decision-making project. She received her PhD in Computer Science from King's College London, UK.

Workshop Format

We propose a half-day workshop, with the following schedule.

10' Workshop welcome and introduction
 30' Keynote / Invited talk
 1h Paper presentations
 30' Coffee break
 1h Paper presentations
 10' Wrap up

Link to Challenges: No

Organisational Requirements: None

Review Policy: Single-anonymous.

We'll use single-anonymous review for submissions to encourage follow-up work, demos, and sharing of tools/resources without the overhead of anonymization.

Publication Policy: Workshop proceedings will be published via CEUR Workshop Proceedings. In case, we are open to collaborating with other workshops to create Joint Proceedings (similar to CEUR-WS Vol-3977).

Program Committee

The PC is composed by the following confirmed members:

- Andrea Sillano, Politecnico di Torino, Italy
- Luigi de Russis, Politecnico di Torino, Italy
- Anisa Rula, University of Brescia, Italy
- Antonio Lieto, University of Salerno, Italy
- Angelo Salatino, The Open University, UK
- Paul Mulholland, The Open University, UK
- Marilena Daquino, University of Bologna, Italy
- Belén Díaz Agudo, Universidad Complutense de Madrid, Spain
- Mark Alan Finlayson, Florida International University, United States
- Enrico Motta, The Open University, UK
- Alessandro Mosca, Free University of Bozen-Bolzano, Italy
- Blerina Spahiu, Università degli Studi di Milano-Bicocca, Italy
- Marieke van Erp, KNAW Humanities Cluster / DHLab, The Netherlands

- Sara Tonelli, Fondazione Bruno Kessler, Italy
- Veruska Zamborlini, KNAW Humanities Cluster / DHLab, The Netherlands
- Rossana Damiano, University of Turin, Italy
- Andrea Schimmenti, University of Bologna, Italy
- Andras Micsik, HUN-REN SZTAKI, Hungary
- Simon Gottschalk, Leibniz Universität Hannover, Germany

Motivation. The rise of Large Language Models (LLMs) has brought both opportunities and challenges in generating coherent, context-aware narratives from complex data. While generative models can produce fluent narratives, they often lack semantic grounding, provenance awareness, and narrative consistency. This workshop is timely as it aims to bridge semantic representations and generative AI, building on established Semantic Web research while addressing current developments in AI. The topic aligns with ongoing special issues on Computational Storytelling at ACM TIST and recent ones in Semantic Web for Cultural Heritage in the Semantic Web Journal, making it highly relevant to ISWC participants interested in explainable, narrative-based access to complex data. Recent publications have shown how ontologies and knowledge graphs can support narrative modelling, analysis, and generation, for example through ontology- or graph-based representations [1,2,6], and semantic infrastructures (mainly KGs) for ensuring the grounding of the generated stories [3,4,5].

Audience. The workshop targets researchers and practitioners working on knowledge graphs, semantic technologies, generative AI, and narrative-based data exploration, including members of the ISWC community active in ontology engineering, Linked Data, reasoning, as well as scholars from NLP, HCI, information retrieval, digital humanities, and creative industries. An active and growing community already exists around these topics, as evidenced by recent works. DDS 2026 is expected to attract both established Semantic Web researchers and contributors exploring the integration of knowledge graphs and GenAI.

Estimated attendance: 30–50 (based on experience at the SEMMES workshop).

Use of Generative AI. ChatGPT was utilized to for grammar-check and rephrasing.

References

1. Pannach et al. (2021), Of lions and Yakshis: Ontology-based narrative structure modelling for culturally diverse folktales, Semantic Web.
2. Meghini et al. (2021), Representing narratives in digital libraries: The narrative ontology, Semantic Web.
3. Plötzky et al. (2024), Lost in Recursion: Mining Rich Event Semantics in Knowledge Graphs, WEBSCI.
4. de Kok et al. (2024), From Nodes to Narratives: A Knowledge Graph-based Storytelling Approach, Text2Story 2024.
5. Wang et al. (2025), Generating Long-form Story Using Dynamic Hierarchical Outlining with Memory-Enhancement, ACL.
6. Chen (2025), Hierarchical Knowledge Graphs for Story Understanding in Visual Narratives, ICIDS 2025.